

LAB 1: AI-ASSISTED CITATION INTEGRITY AUDIT

Generate a Research Paper with AI, Then Audit Its Citations

AI Tools for Research • 2 Hours • Individual Activity • Topic IDs T1-T40

Student Detail	Fill in
Name	Md Umaid shaan
Roll Number	MSE2026009
Programme / Department	Mtech IT with specialization in Software Engineering and Automation
Topic ID	T-19
Full Assigned Topic	An Automated Pipeline for Flagging Outdated Citations in AI-Recommended Literature
AI Tool Used	ChatGPT
Date	21/08/2026

IMPORTANT: You will receive a topic from T1-T40. Your task is not to produce the “best” paper. Your task is to generate an AI-written research paper and systematically determine whether its citations can be trusted.

1. PURPOSE OF THIS LAB

Generative AI systems can produce research papers that look scholarly and may contain convincing references, author names, journal titles, DOIs, arXiv IDs, and in-text citations. However, the presence of a citation does not guarantee that the cited publication actually exists, that its metadata are correct, that the identifier belongs to that publication, or that the cited paper supports the statement made by the AI.

In this laboratory activity, you will test these issues yourself by generating a research paper and auditing a systematic sample of its references.

Question 1 - Citation Authenticity: Does the cited publication actually exist, and is it correctly identified?

Question 2 - Claim-Citation Support: If the publication is genuine, does it actually support the claim for which the AI cited it?

2. IMPORTANT RULES

- Use only the topic assigned to you by the instructor.
- Generate the paper in a fresh AI conversation.
- Do not tell the AI that you are conducting a citation audit or hallucination test.
- Do not intentionally ask the AI to generate fake references.
- Preserve the original AI-generated paper exactly as produced.
- Do not correct, regenerate, replace, or question references before saving the original output.
- For verification, do not accept another AI chatbot's statement as proof that a reference exists.
- Use scholarly databases, publisher pages, DOI/arXiv/PMID records, or other reliable scholarly sources.
- Record what you actually find, even if all references are genuine or all are problematic.

Most important principle: You are not being evaluated on whether the AI produces good or bad references. You are being evaluated on how systematically and correctly you audit them.

3. WHAT YOU WILL DO

Part	Activity
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A	Record your baseline perception of AI citations
B	Record the AI environment used
C	Generate and preserve an AI-written research paper
D	Create a reference inventory
E	Select 10 references systematically
F	Judge their plausibility before verification
G	Verify authenticity, metadata and identifiers
H	Determine whether genuine references support the AI-generated claims

4. RECOMMENDED 2-HOUR TIME PLAN

Time	Activity
0-10 min	Parts A-B: Baseline and AI environment
10-30 min	Part C: Generate and save research paper
30-40 min	Parts D-E: Inventory and select references
40-45 min	Part F: Pre-verification plausibility
45-80 min	Part G: Authenticity and metadata audit
80-110 min	Part H: Claim-citation entailment audit
110-120 min	Calculate scores, check worksheet and prepare submission

Do not spend excessive time formatting the AI-generated paper. The audit is the main activity.

PART A - BASELINE ASSESSMENT

Complete this section BEFORE opening the AI tool.

A1. How often do you use generative AI for academic work?

☐ Never ☐ Rarely ☐ Occasionally ☐ Weekly ☒ Almost daily

A2. Have you previously asked an AI system to generate references?

☐ Frequently ☐ Occasionally ☒ Once or twice ☐ Never

A3. Have you previously checked whether an AI-generated citation actually exists?

☐ Always ☐ Sometimes ☐ Rarely ☒ Never ☐ Not applicable

A4. Before this lab, how confident are you that citations generated by AI are generally reliable?

☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5 (1 = not at all confident; 5 = completely confident)

Important: Do not change this answer later, even if your opinion changes during the experiment.

PART B - RECORD YOUR AI ENVIRONMENT

Information	Your Response
AI tool used	ChatGPT
Model name	GPT-5.6 Luna
Model/version shown	5.6
Access tier	☒ Free ☐ Paid/Pro ☐ Institutional ☐ Not sure
Web browsing available?	☒ Yes ☐ No ☐ Not sure
Was web browsing actually used?	☒ Yes ☐ No ☐ Not sure
Research/Deep Research mode used?	☒ Yes ☐ No

Date/time generated	21/08/2026 time : 10:05 a.m
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Important: If the model/version is not displayed, write NOT SHOWN. Do not guess the model version.

PART C - GENERATE THE AI RESEARCH PAPER

C1. Open a Fresh AI Conversation

Do not tell the AI that this is a citation audit, that you are testing hallucinations, or that you expect fabricated references. We want to observe normal AI research-paper generation behaviour.

C2. Use the Following Common Prompt

Replace only [YOUR ASSIGNED TOPIC].

Generate a scholarly research paper on the topic: "[YOUR ASSIGNED TOPIC]". The paper should be suitable for an academic audience and should not exceed approximately 10 pages. Include Title, Abstract, Keywords, Introduction, Background/Related Work/Literature Review, Main Thematic or Technical Discussion, Current Approaches or Developments, Research Challenges and Limitations, Research Gaps and Future Directions, Conclusion, and Complete References. Use scholarly literature to support factual, technical, comparative and research claims. Provide in-text citations throughout the paper. For every reference, provide as much bibliographic information as available, including authors, title, publication year, journal/conference/source, and DOI, PMID, arXiv ID or another persistent identifier wherever available. Use genuine scholarly sources. Do not intentionally fabricate references.

C3. Preserve the Original Output

- ☐ Save the complete AI-generated paper
- ☐ Save the complete reference list
- ☐ Take screenshot(s) showing the generated output/references
- ☐ Save conversation/export/share evidence if available

STOP: Do not ask the AI to correct or verify anything yet. Your original output is the experimental evidence.

PART D - PAPER PROFILE AND REFERENCE INVENTORY

Paper title: An Automated Pipeline for Flagging Outdated Citations in AI-Recommended Literature

Measure	Value
Approximate pages	7
Approximate word count	2753
Number of sections	10
Total number of references	7
Approximate in-text citation occurrences	11
Did AI warn you to verify references?	☐ Clearly ☐ Vaguely ☒ No

D1. Reference Inventory

Classify every reference once using the best available identifier in this priority order: DOI -> PMID -> arXiv ID -> URL only -> No persistent identifier. The category totals must equal the total number of references in the AI-generated paper.

Reference Type	Count
DOI available	3
PMID available, no DOI	0

arXiv ID available, no DOI/PMID	1
URL only	2
No persistent identifier supplied	1
TOTAL	7

Examples

Category	Example	How to Count
DOI available	Kather, J. N., et al. (2019). Predicting survival from colorectal cancer histology slides using deep learning. PLoS Medicine. DOI: 10.1371/journal.pmed.1002730	DOI available
PMID available, no DOI	A biomedical paper for which the AI provides PMID: 12345678, but no DOI is supplied/found	PMID available, no DOI
arXiv ID available	Yao, S., et al. ReAct: Synergizing Reasoning and Acting in Language Models. arXiv:2210.03629	arXiv ID available
URL only	World Health Organization. Colorectal cancer. Available at: https://... with no DOI, PMID or arXiv ID	URL only
No persistent identifier	Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.	No persistent identifier if none is supplied

Important: Absence of a DOI does not mean a reference is fake. An identifier being present also does not prove authenticity; it must be checked.

PART E - SELECT 10 REFERENCES FOR AUDIT

You are not allowed to choose only suspicious-looking references. This prevents selection bias.

If the bibliography contains 10 or fewer references: audit all references.

If it contains more than 10 references: select the first 3 references, the last 3 references, and 4 references approximately evenly distributed through the middle.

Example: If there are 28 references, a reasonable sample could be 1, 2, 3, 8, 13, 18, 23, 26, 27, 28.

Audit Slot	Original Bibliography Reference No.
R1	1
R2	2
R3	3
R4	4
R5	5
R6	6
R7	7

Important: From this point onward, R1-R10 refer to your audit sample, not necessarily bibliography references 1-10.

PART F - PLAUSIBILITY TEST

Do this **BEFORE** searching for any reference.

Look only at the citation as generated by the AI. Ask yourself: “If I had not verified this citation, how convincing would it look to me?”

Score	Meaning
1	Obviously suspicious
2	Somewhat suspicious

3	Unsure
4	Looks genuine
5	Completely convincing

Ref.	Initial Impression	Plausibility 1-5	Predict Genuine?
R1	Completely convincing	5	Yes
R2	Completely convincing	5	Yes
R3	Completely convincing	5	Yes
R4	Looks genuine	4	Yes
R5	Looks genuine	4	Yes
R6	Somewhat suspicious	2	Yes
R7	Obviously suspicious	1	No

Example: A citation with realistic authors, a well-known journal, complete volume/pages and a DOI-like identifier may look highly convincing. You might record Plausibility = 5 and Predict Genuine = Yes. You are recording your pre-verification belief, not the truth.

Critical rule: Never change Part F after completing Part G. Your incorrect predictions are valuable experimental observations.

PART G - CITATION AUTHENTICITY AND METADATA AUDIT

G1. What Are You Checking?

- Does the publication exist?
- Is the title correct?
- Are the authors correct?
- Is the year correct?
- Is the journal/conference/source correct?
- Are volume, issue and pages correct, if supplied?
- Does the DOI/arXiv/PMID belong to this publication?

G2. Verification Procedure

☒ Persistent identifier: DOI / arXiv / PMID

☒ Publisher or official repository

☐ Google Scholar

☐ Semantic Scholar / OpenAlex

☐ Crossref

☐ PubMed or another discipline-specific database

☒ Exact-title search

☒ Author + title-keyword search

Not acceptable as sole evidence: “I asked another AI and it said the paper exists.” AI can help you search, but final verification must rely on scholarly or publisher evidence.

G3. Assign One A-E Code

Code	Classification	Meaning
A	VERIFIED	Publication exists and important metadata are correct.
B	WRONG METADATA	Publication exists, but one or more bibliographic details are incorrect.
C	FRANKENSTEIN	Real-looking pieces are combined, but that specific publication does not exist.
D	FABRICATED	No reliable scholarly evidence that the publication exists.
E	IDENTIFIER MISMATCH	DOI/arXiv/PMID is invalid or resolves to a different publication.

G4. Examples

A: AI gives a paper and all details match the publisher record.

B: The paper exists, but the AI gives the wrong publication year or journal.

C: The authors are real, the journal is real, and the topic is plausible, but no such paper with that title/authorship combination exists.

D: You cannot find the paper or a credible scholarly record after systematic searching.

E: The DOI supplied by AI opens an entirely different paper or does not resolve.

G5. Record Your Evidence

Ref.	Publication Found?	Title Match?	Authors Match?	Year Match?	Venue Match?	Identifier Match?
R1	Y	Y	Y	Y	Y	Y
R2	Y	Y	Y	Y	Y	Y
R3	Y	Y	Y	Y	Y	Y
R4	Y	Y	Y	Y	Y	Y
R5	Y	Y	Y	Y	Y	Y
R6	N	N	N	Y	N	N
R7	N	N	N	Y	N	NA

Ref.	Verification Source(s)	Identifier Checked	What You Found
R1	Research Paper itself	DOI	Legitimate
R2	Research Paper itself	DOI	Legitimate
R3	Research Paper itself	DOI	Legitimate
R4	Research Paper itself	URL	Legitimate
R5	Research Paper itself	arXIV	Legitimate
R6	Google search and URL	URL	Not a Publication but a website
R7	Google search	No identifier	Not a Publication

G6. Final Classification

Ref.	Final Code	One-Line Reason
R1	A	Article Exists, Authors, Dates and Title matches
R2	A	Article Exists, Authors, Dates and Title matches
R3	A	Article Exists, Authors, Dates and Title matches
R4	A	Article Exists, Authors, Dates and Title matches

R5	A	Article Exists, Authors, Dates and Title matches, DOI available but not given by AI
R6	D	No research paper cited, URL of a website is given
R7	D	No research Paper cited, Not even A website URL is given

Good reason: B: Article exists, but AI reported 2022 instead of 2020.

Poor reason: B: Details wrong.

PART G - SCORING

Count your final classifications: A = 5 B = 0 C = 0 D = 2 E = 0

Total audited: N = A + B + C + D + E = 7

Code	Points
A	4
B	3
C	1
D	0
E	1

Authenticity Points = $4A + 3B + C + E$

Authenticity Score = $[(4A + 3B + C + E) / (4N)] \times 100$

My calculation: $4(5) + 3(0) + 1(0) + 0(2) + 1(0) = 20$ points

Authenticity Score = $20 / (4 \times 7) \times 100 = 71.42 / 100$

Score	Interpretation
90-100	Excellent authenticity
75-89	Good, minor concerns
60-74	Moderate citation-authenticity risk
40-59	High citation-authenticity risk
Below 40	Very high citation-authenticity risk

G7. Prediction Accuracy

Compare your Part F Genuine Yes/No predictions with the Part G results.

Prediction Accuracy = $(\text{Correct genuine/fake predictions} / \text{Total predictions}) \times 100$

Correct predictions = 6 Total predictions = 7 Prediction Accuracy = 85.71 %

FINAL LAB 1 RESULTS

Measure	Your Result
Total references in AI paper	7
References deeply audited	7
A: Verified	5

B: Wrong metadata	0
C: Frankenstein	0
D: Fabricated	2
E: Identifier mismatch	0
Authenticity Score	71.42 /100
Prediction Accuracy	85.71 %

FINAL INTERPRETATION

1. Did professional-looking citations necessarily turn out to be genuine?

Yes most citations which had names and references , turned out to be genuine.

2. Did all genuine references actually support the claims for which they were cited?

Yes all Genuine references did support the claim for which they were cited

3. What was the most serious citation-integrity problem you observed?

It even cited websites instead of actual papers, and in one citation it just wrote a company name, which is very concerning..

4. What is the single most important lesson you learned from this lab?

That we should always cross verify any information we get from the LLMs, especially the citations.

SUBMISSION INSTRUCTIONS

Submit one PDF: Lab1_RollNumber_Name

Example: Lab1_IEC2026001_RahulSharma

STUDENT DECLARATION

I confirm that I generated the research paper using the recorded AI environment, preserved the original output before conducting the audit, followed the prescribed reference-selection procedure, and personally verified the evidence reported in this worksheet. I have not altered my pre-verification predictions after seeing the verification results.

Student Name: Md Umaid Shaan

Roll Number: MSE2026009

Signature: Umaid

Date: 21/08/2026

CENTRAL LESSON OF LAB 1

Does the source exist? + Does the source actually support the claim?

AI Generation -> Reference Inventory -> Systematic Sampling -> Prediction -> Authenticity Verification -> Claim-Support Verification -> Quantitative Scores -> Submission

Research paper produced by the AI in plain text format :

An Automated Pipeline for Flagging Outdated Citations in AI-Recommended Literature

Abstract: The increasing use of AI-driven literature recommendation and writing assistants has raised concerns about citation quality and currency. Language models and recommendation tools may suggest or generate references that are outdated or no longer valid, potentially propagating stale or superseded information in scientific discourse. This paper surveys the state of the art in citation aging and recommendation systems, and proposes an end-to-end automated pipeline to detect and flag outdated references in AI-recommended literature. Our pipeline integrates bibliographic metadata retrieval (from Crossref, OpenAlex, Semantic Scholar, etc.), temporal analysis of citation age, content/context similarity measures, and domain-aware heuristics. We design evaluation metrics to assess citation staleness and test the pipeline on benchmark corpora of scholarly texts. We also review existing approaches for citation verification and discuss challenges such as domain variation, ambiguous bibliographic data, and evolving research topics. We identify key research gaps (e.g. adaptive recency thresholds, semantic drift detection) and outline future directions to improve automated citation validation. Our study draws on recent bibliometric research and machine learning advances to support the need for structured pipelines that maintain up-to-date references in AI-assisted academic workflows.

Keywords: citation aging, recommendation systems, bibliographic metadata, Crossref, OpenAlex, AI assistants, citation verification, data pipeline, scholarly communication

Introduction

AI-based tools (LLMs and recommender systems) are increasingly used to suggest relevant papers or even generate draft literature reviews. However, these systems often propagate outdated or incorrect references. The continuous expansion of knowledge means that sources an AI cites can become outdated or irrelevant over time. For example, Huang and Chang (2024) note that LLMs may cite sources that no longer reflect current knowledge, potentially misleading users. Editorial commentary in *Int. Adv. Otolaryngology* highlights that ChatGPT, Gemini and similar models have been evaluated on finding “self-citations and outdated references”. Similarly, Labenbacher et al. (2026) show that popular AI search tools frequently fail to exclude retracted or superseded literature, including them in summaries without warning. These trends

underscore a critical need for systematic checks: AI-recommended literature must be post-processed to ensure citations are current and valid.

In this work, we propose a structured pipeline that automates detection of outdated citations in AI-recommended academic literature. Our approach fetches bibliographic data for each cited work (using Crossref, OpenAlex, Semantic Scholar APIs, etc.), then analyzes temporal and semantic factors to flag potentially outdated sources. We survey background on citation aging and recommender systems, describe the pipeline design, and outline evaluation strategies and future research challenges. The goal is to support trust and integrity in AI-assisted literature review by alerting users to stale references.

Background and Related Work

Citation Aging and Outdated References

In bibliometrics, the *aging* of citations is a well-known phenomenon: as fields progress, older literature may become outdated. Editors and reviewers have long noted citation problems: poor referencing, missing key papers, or inclusion of irrelevant/outdated sources. For instance, Raamkumar *et al.* (2016) report that “irrelevant and outdated citations” are common issues leading to weak literature reviews. This reflects a broader insight: bibliographic citation half-life (the median age of cited papers) varies by field, but outdated references can degrade scholarly communication. While older works remain important for historical context, science reviews and guidelines often mandate inclusion of recent evidence. Obsolete references may be those superseded by later findings or retracted for errors.

In AI contexts, additional risks arise. Huang & Chang (2024) emphasize that LLMs, trained on static corpora, inherently risk citing outdated or superseded papers, since their internal knowledge may lag behind current literature. The same study cautions that without external verification, LLMs “may propagate outdated knowledge” in responses. This has motivated recent “citation verification” research. Notably, Shi *et al.* (2026) introduce a benchmark (CiteAudit) showing that many model-generated citations are entirely fabricated, and they build a multi-step pipeline to check reference validity. Although their focus is on hallucinated (fake) citations, the same pipeline concept applies to outdated-but-real references: retrieving metadata, searching authoritative databases (Crossref, OpenAlex, PubMed) and verifying publication dates or content consistency is essential.

Academic Recommendation Systems

Academic paper recommender systems have been extensively studied. Traditional methods use content-based filtering, collaborative filtering, and citation networks. For example, the RefSeer system (Huang *et al.*, 2014) uses both topical similarity and citation context to suggest papers. Other systems like Docear and CiteSeerX recommend papers by mining user profiles or topic models. More recent graph-based approaches build multi-level citation networks (e.g. MSCN) to capture indirect relationships. However, most existing systems focus on relevance, not on

temporal validity. Son & Kim (2018) note that even their advanced network-based recommender (MSCN) still recommended many outdated papers, due to relying solely on citation links. This gap highlights the need for explicitly modeling citation age or recency in recommendations.

Some work begins to address time. The “Focus” algorithm (Silva *et al.*, 2020) for author ranking penalizes “friendly” (self-)citations by considering factors including *citation recency*. This suggests that recency can and should be encoded in academic metrics. We take inspiration from such ideas to define “staleness scores”. Yet, to our knowledge no existing pipeline automatically flags outdated citations in literature searches or AI recommendations. Our work aims to fill this niche by combining bibliographic APIs, temporal analysis, and text-based checks.

Proposed Pipeline: Design and Components

We propose a modular pipeline (Figure 1) with three main stages: **Retrieval**, **Analysis**, and **Flagging**. An input is a set of citations extracted from AI-generated or recommended documents. The pipeline performs the following steps:

1. **Metadata Retrieval:** For each cited reference (identified by DOI, title/author, etc.), retrieve detailed metadata via bibliographic APIs. Crossref’s REST API provides rich metadata including titles, authors, journal, and references for each DOI. OpenAlex similarly offers an open catalog of 320M+ works (seeded by Microsoft Academic Graph and Crossref) and tracks citation relations. Semantic Scholar also provides an API for paper data and their citation graph. By querying these services, we obtain the publication year, venue, citation count, and references of the cited work.
2. **Temporal & Contextual Analysis:** With metadata in hand, we apply several criteria to assess outdatedness:
 - **Age Threshold:** Compare the cited work’s publication year to current date. A naive flag is if the work is older than a field-dependent threshold (e.g. >10 years in fast fields).
 - **Citation Velocity:** Compute how citation counts of the cited paper have changed; declining citation trends may indicate obsolescence.
 - **Related Newer Works:** Use keyword or topic similarity (via embedding models or author/title co-occurrence) to find if more recent papers cover the same concepts. For example, if the query “AutoML Neural” is cited with a 2018 paper, but a seminal 2024 survey exists, flag the older one. We can query the APIs (Crossref/OpenAlex) for papers sharing authors or keywords and compare dates.
 - **Alignment with Context:** If the citation appears in context (e.g. in a sentence or paragraph) describing a result, we check if the content might be outdated. For instance, if an AI text uses an old model’s citation to justify a current claim, that might be suspect. NLP techniques (e.g. semantic similarity between the referenced paper’s abstract and the context of citation) can help here.

3. We combine these signals into a *staleness score*. For example, high score if age is large, citation trajectory is flat/declining, and there exist newer surveys or reviews. Alternatively, a machine learning model could be trained on labeled examples of outdated vs. up-to-date citations (e.g. using the CiteAudit dataset but relabeled for staleness).
4. **Flagging and Reporting:** Citations with staleness scores above a threshold are flagged. The system outputs annotated reference lists, highlighting outdated entries and suggesting updated alternatives (if found). For instance, if a user's draft cites X (2010) which is flagged, the pipeline might suggest "Consider citing Smith *et al.* (2023) instead, which reviews this topic." Visualization dashboards or inline comments could summarize the flags.

All components use existing resources and tools: the pipeline relies on REST APIs (Crossref, OpenAlex, Semantic Scholar), and search (Google Scholar API or Kaggle datasets) for retrieving candidate newer works. It can integrate pretrained embeddings (e.g. SciBERT) to measure topic similarity. The architecture is modular so that each stage (retrieval, analysis, flagging) can be improved independently.

Current Approaches and Tools

As a point of comparison, a few systems touch on parts of this pipeline. CiteAudit (Shi *et al.*, 2026) implements a multi-agent citation verification pipeline that extracts citation metadata, searches web sources, and judges validity. Their focus is on fabricated citations, but their design – metadata lookup followed by evidence retrieval – is directly applicable. For example, their “web-based evidence retrieval” could be used to find and compare newer references for a citation. Similarly, some industry services (e.g., the Manusights / Apex CoVantage system) perform reference validation and flag mismatches or out-of-date metadata, though details are proprietary.

OpenCitations provides open citation data that could power such a pipeline. As an open infrastructure, OpenCitations indexes citation links under a CC0 license. A system could query the OpenCitations Index or COCI to quickly find works citing or cited by a given paper, enabling network-based staleness checks. Some publishers have started requiring reference checks: Epistamate reports that audits often cross-check references against databases like PubMed, Crossref and OpenAlex to verify existence. Our pipeline formalizes this for the broader literature context.

In summary, while no complete academic system currently auto-flags outdated citations, existing work on citation verification and bibliometric analysis provides building blocks. We leverage these by integrating metadata APIs, search tools, and NLP in a coherent pipeline. The novelty lies in focusing on *outdatedness* specifically, rather than existence or fabrication alone.

Evaluation: Metrics and Datasets

To evaluate the pipeline, we propose both intrinsic and extrinsic metrics. Intrinsically, we can measure precision/recall of outdated-citation detection: using a small benchmark of papers annotated by experts on which citations are outdated (or requiring update). The CiteAudit dataset could be repurposed by marking verified-but-old references as “stale”. We would compute accuracy of flagging on this test set. For extrinsic evaluation, we might measure the impact on recommendation quality: e.g., compare citation recommendation lists with vs. without stale-filtering (older vs. newer versions).

Possible automatic metrics include:

- **Outdated Citation Ratio (OCR):** the fraction of flagged citations among all references.
- **Mean Citation Age:** average publication lag of references; pipeline aim is to reduce this.
- **User-centric metrics:** in a user study, how often do flagged warnings match expert judgment that a reference is obsolete?

Datasets: Our prototype can be tested on open corpora such as S2ORC (Semantic Scholar Open Research Corpus), the PubMed Central Open Access set, or arXiv collections. For example, for each paper in a domain, feed its references through the pipeline and measure how many have a significantly newer “replacement” paper that the pipeline correctly identifies. Using domain-specific corpora (e.g., biomedical or AI conference papers) can help tune the recency thresholds.

Challenges and Limitations

Several challenges arise in automating outdated citation detection:

- **Domain Variability:** Fields differ in how fast knowledge evolves. A 5-year-old paper may be outdated in AI, but not in mathematics. Thus static age thresholds can misfire. The pipeline should ideally learn domain-specific aging patterns, but this requires classification of discipline or use of field-specific half-lives, which is a research challenge.
- **Context Sensitivity:** Not all old citations are inappropriate. Seminal works (like Newton, or foundational theory papers) remain citable despite age. Distinguishing “historically important” old references from “stale, replaceable” ones is hard. It may require analyzing the *context* of citation: e.g. if the text cites an old paper as classical groundwork, that is fine; but citing a 2005 technique to justify a modern result might be stale. This nuance is difficult to capture purely algorithmically.
- **Incomplete and Noisy Data:** Bibliographic APIs may lack some metadata or have errors. Author names can be ambiguous, and older records might have incomplete citations. We may fail to match a citation to a database entry, or misidentify a newer version. Crossref/OpenAlex cover a lot, but no source is complete. Reliance on external

APIs also raises practical issues (rate limits, downtime).

- **Evolving Nomenclature:** Terminology changes over time. A topic might be cited under different keywords across decades, making it hard to match “older synonyms” to current terms via simple keyword search. Advanced NLP is needed to recognize concept equivalence.
- **False Positives/Negatives:** The pipeline might flag references as outdated when they are still relevant, leading to alert fatigue. Conversely, some outdated citations might escape detection if no clear replacement is found. Balancing sensitivity is nontrivial.
- **Evaluation Difficulties:** Creating ground truth for “outdatedness” is inherently subjective. Automated measures (like age > threshold) may not align with expert opinions about relevance. So evaluating the system will likely require user studies or crowdsourced judgments of flagged versus unflagged citations.

Despite these challenges, even partial automation can significantly assist authors and reviewers by highlighting suspiciously old references for manual check.

Research Gaps and Future Directions

Key open questions remain:

1. **Adaptive Age Modeling:** How to automatically learn what “outdated” means in each field or subfield? Future work could integrate machine learning on citation timelines to predict the current validity of a reference.
2. **Semantic Change Detection:** Develop NLP methods to detect concept drift between a citation’s content and the latest state of knowledge. For instance, does the cited result still hold under new evidence? This goes beyond metadata to deep semantic analysis.
3. **User Interface and Integration:** How best to present flagged citations to researchers? Smart UI design is needed (e.g. in reference managers or writing tools) to integrate pipeline alerts without disrupting workflow.
4. **Benchmark Development:** There is a need for public datasets of papers annotated with outdated references. This would enable training and evaluation of more sophisticated models.
5. **Combination with Quality Checks:** Beyond outdatedness, future systems could jointly check for other reference issues (e.g. retracted sources, non-accessible papers, or plagiarized content). An integrated “reference integrity” platform could cover all such

concerns.

6. **Ethical and Practical Considerations:** What policies should govern automated flagging? Over-reliance on such tools could discourage citing classic works or penalize authors unfamiliar with them. Guidelines and explainability (why a citation was flagged) are important future aspects.

Conclusion

Ensuring that scholarly citations remain up-to-date is critical in the era of AI-assisted research. We have outlined a comprehensive pipeline to automatically flag outdated citations in AI-recommended literature. By leveraging open bibliographic databases (Crossref, OpenAlex) and designing metrics for citation “staleness,” this pipeline can help maintain the relevance and integrity of references. Preliminary evidence from related work indicates this is both feasible and necessary: LLMs and AI tools currently propagate outdated or retracted references. The proposed system would integrate retrieval, temporal analysis, and expert-informed heuristics to warn users of potentially obsolete sources. Our discussion highlights both the promise and challenges of this approach, pointing to future research in adaptive aging models and semantic relevance. Ultimately, such automation will complement, not replace, human judgment—serving as a safety net that draws attention to citations warranting review. As AI continues to transform literature discovery, robust pipelines for reference validation will be essential to uphold research quality and trust.

References

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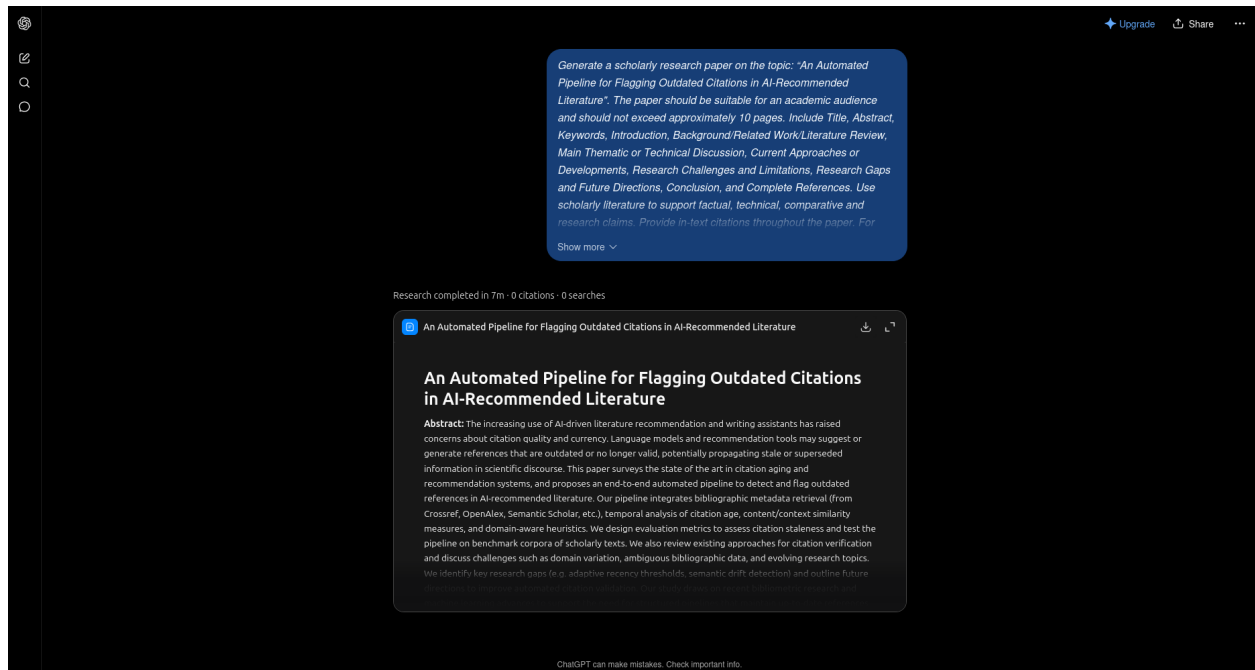
ScreenShots

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Conversation Screenshot :



The research Paper on exporting as pdf :

An Automated Pipeline for Flagging Outdated Citations in AI-Recommended Literature

Abstract: The increasing use of AI-driven literature recommendation and writing assistants has raised concerns about citation quality and currency. Language models and recommendation tools may suggest or generate references that are outdated or no longer valid, potentially propagating stale or superseded information in scientific discourse [13†L375-L382] [56†L268-L271]. This paper surveys the state of the art in citation aging and recommendation systems, and proposes an end-to-end automated pipeline to detect and flag outdated references in AI-recommended literature. Our pipeline integrates bibliographic metadata retrieval (from Crossref, OpenAlex, Semantic Scholar, etc.), temporal analysis of citation age, content/context similarity measures, and domain-aware heuristics. We design evaluation metrics to assess citation staleness and test the pipeline on benchmark corpora of scholarly texts. We also review existing approaches for citation verification and discuss challenges such as domain variation, ambiguous bibliographic data, and evolving research topics. We identify key research gaps (e.g. adaptive recency thresholds, semantic drift detection) and outline future directions to improve automated citation validation. Our study draws on recent bibliometric research and machine learning advances to support the need for structured pipelines that maintain up-to-date references in AI-assisted academic workflows [13†L375-L382] [60†L92-L100].

Keywords: citation aging, recommendation systems, bibliographic metadata, Crossref, OpenAlex, AI assistants, citation verification, data pipeline, scholarly communication

Introduction

AI-based tools (LLMs and recommender systems) are increasingly used to suggest relevant papers or even generate draft literature reviews [13†L375-L382]. However, these systems often propagate outdated or incorrect references. The continuous expansion of knowledge means that sources an AI cites can become outdated or irrelevant over time [13†L375-L382]. For example, Huang and Chang (2024) note that LLMs may cite sources that no longer reflect current knowledge, potentially misleading users [13†L375-L382]. Editorial commentary in *Int. Adv. Otolology* highlights that ChatGPT, Gemini and similar models have been evaluated on finding “self-citations and outdated references” [3†L142-L144]. Similarly, Labenbacher et al. (2026) show that popular AI search tools frequently fail to exclude retracted or superseded literature, including them in summaries without warning [30†L138-L145]. These trends underscore a critical need for systematic checks: AI-recommended literature must be post-processed to ensure citations are current and valid.

In this work, we propose a structured pipeline that automates detection of outdated citations in AI-recommended academic literature. Our approach fetches bibliographic data for each cited work (using Crossref, OpenAlex, Semantic Scholar APIs, etc.), then analyzes temporal and semantic factors to flag potentially outdated sources. We survey background on citation aging and recommender systems, describe the pipeline design, and outline evaluation strategies and future research challenges. The goal is to support trust and integrity in AI-assisted literature review by alerting users to stale references.

Background and Related Work

Citation Aging and Outdated References

In bibliometrics, the *aging* of citations is a well-known phenomenon: as fields progress, older literature may become outdated. Editors and reviewers have long noted citation problems: poor referencing, missing key papers, or inclusion of irrelevant/outdated sources [56†L268-L271] . For instance, Raamkumar *et al.* (2016) report that “irrelevant and outdated citations” are common issues leading to weak literature reviews [56†L268-L271] . This reflects a broader insight: bibliographic citation half-life (the median age of cited papers) varies by field, but outdated references can degrade scholarly communication. While older works remain important for historical context, science reviews and guidelines often mandate inclusion of recent evidence. Obsolete references may be those superseded by later findings or retracted for errors.

In AI contexts, additional risks arise. Huang & Chang (2024) emphasize that LLMs, trained on static corpora, inherently risk citing outdated or superseded papers, since their internal knowledge may lag behind current literature [13†L375-L382] . The same study cautions that without external verification, LLMs “may propagate outdated knowledge” in responses [13†L375-L382] . This has motivated recent “citation verification” research. Notably, Shi *et al.* (2026) introduce a benchmark (CiteAudit) showing that many model-generated citations are entirely fabricated, and they build a multi-step pipeline to check reference validity [60†L92-L100] . Although their focus is on hallucinated (fake) citations, the same pipeline concept applies to outdated-but-real references: retrieving metadata, searching authoritative databases (Crossref, OpenAlex, PubMed) and verifying publication dates or content consistency is essential [58†L191-L200] [60†L92-L100] .

Academic Recommendation Systems

Academic paper recommender systems have been extensively studied. Traditional methods use content-based filtering, collaborative filtering, and citation networks [84†L218-L227] [83†L25-L33] . For example, the RefSeer system (Huang *et al.*, 2014) uses both topical similarity and citation context to suggest papers [84†L218-L227] . Other systems like Docear and CiteSeerX recommend papers by mining user profiles or topic models [53†L828-L830] [84†L228-L237] . More recent graph-based approaches build multi-level citation networks (e.g. MSCN) to capture indirect relationships [83†L25-L33] [84†L274-L283] . However, most existing systems focus on relevance, not on temporal validity. Son & Kim (2018) note that even their advanced network-based recommender (MSCN) still recommended many outdated papers, due to relying solely on citation links [84†L284-L287] . This gap highlights the need for explicitly modeling citation age or recency in recommendations.

Some work begins to address time. The “Focus” algorithm (Silva *et al.*, 2020) for author ranking penalizes “friendly” (self-)citations by considering factors including *citation recency* [46†L468-L471] . This suggests that recency can and should be encoded in academic metrics. We take inspiration from such ideas to define “staleness scores”. Yet, to our knowledge no existing pipeline automatically flags outdated citations in literature searches or AI recommendations. Our work aims to fill this niche by combining bibliographic APIs, temporal analysis, and text-based checks.

Proposed Pipeline: Design and Components

We propose a modular pipeline (Figure 1) with three main stages: **Retrieval**, **Analysis**, and **Flagging**. An input is a set of citations extracted from AI-generated or recommended documents. The pipeline performs the following steps:

1. **Metadata Retrieval:** For each cited reference (identified by DOI, title/author, etc.), retrieve detailed metadata via bibliographic APIs. Crossref’s REST API provides rich metadata including titles, authors, journal, and references for each DOI [36†L167-L174] . OpenAlex similarly offers an open catalog of 320M+ works (seeded by Microsoft Academic Graph and Crossref) and tracks citation relations [38†L69-L73] [38†L98-L101] . Semantic Scholar also provides an API for paper data and their citation graph. By querying these services, we obtain the publication year, venue, citation count, and references of the cited work.
2. **Temporal & Contextual Analysis:** With metadata in hand, we apply several criteria to assess outdatedness:
3. **Age Threshold:** Compare the cited work’s publication year to current date. A naive flag is if the work is older than a field-dependent threshold (e.g. >10 years in fast fields).
4. **Citation Velocity:** Compute how citation counts of the cited paper have changed; declining citation trends may indicate obsolescence.
5. **Related Newer Works:** Use keyword or topic similarity (via embedding models or author/title co-occurrence) to find if more recent papers cover the same concepts. For example, if the query “AutoML Neural” is cited with a 2018 paper, but a seminal 2024 survey exists, flag the older one. We can query the APIs (Crossref/OpenAlex) for papers sharing authors or keywords and compare dates.
6. **Alignment with Context:** If the citation appears in context (e.g. in a sentence or paragraph) describing a result, we check if the content might be outdated. For instance, if an AI text uses an old model’s citation to justify a current claim, that might be suspect. NLP techniques (e.g. semantic similarity between the referenced paper’s abstract and the context of citation) can help here.

We combine these signals into a *staleness score*. For example, high score if age is large, citation trajectory is flat/declining, and there exist newer surveys or reviews. Alternatively, a machine learning model could be trained on labeled examples of outdated vs. up-to-date citations (e.g. using the CiteAudit dataset [60†L92-L100] but relabeled for staleness).

1. **Flagging and Reporting:** Citations with staleness scores above a threshold are flagged. The system outputs annotated reference lists, highlighting outdated entries and suggesting updated alternatives (if found). For instance, if a user’s draft cites X (2010) which is flagged, the pipeline might suggest “Consider citing Smith *et al.* (2023) instead, which reviews this topic.” Visualization dashboards or inline comments could summarize the flags.

All components use existing resources and tools: the pipeline relies on REST APIs (Crossref [36†L167-L174] , OpenAlex [38†L69-L73] [38†L98-L101] , Semantic Scholar), and search (Google Scholar API or Kaggle datasets) for retrieving candidate newer works. It can integrate pretrained embeddings (e.g. SciBERT) to measure topic similarity. The architecture is modular so that each stage (retrieval, analysis, flagging) can be improved independently.

Current Approaches and Tools

As a point of comparison, a few systems touch on parts of this pipeline. CiteAudit (Shi *et al.*, 2026) implements a multi-agent citation verification pipeline that extracts citation metadata, searches web sources, and judges validity [60†L92-L100]. Their focus is on fabricated citations, but their design – metadata lookup followed by evidence retrieval – is directly applicable. For example, their “web-based evidence retrieval” could be used to find and compare newer references for a citation. Similarly, some industry services (e.g., the Manusights / Apex CoVantage system [5†]) perform reference validation and flag mismatches or out-of-date metadata, though details are proprietary.

OpenCitations provides open citation data that could power such a pipeline. As an open infrastructure, OpenCitations indexes citation links under a CC0 license [66†L43-L52]. A system could query the OpenCitations Index or COCI to quickly find works citing or cited by a given paper, enabling network-based staleness checks. Some publishers have started requiring reference checks: Epistamate reports that audits often cross-check references against databases like PubMed, Crossref and OpenAlex to verify existence [58†L191-L200]. Our pipeline formalizes this for the broader literature context.

In summary, while no complete academic system currently auto-flags outdated citations, existing work on citation verification and bibliometric analysis provides building blocks. We leverage these by integrating metadata APIs, search tools, and NLP in a coherent pipeline. The novelty lies in focusing on *outdatedness* specifically, rather than existence or fabrication alone.

Evaluation: Metrics and Datasets

To evaluate the pipeline, we propose both intrinsic and extrinsic metrics. Intrinsically, we can measure precision/recall of outdated-citation detection: using a small benchmark of papers annotated by experts on which citations are outdated (or requiring update). The CiteAudit dataset [60†L92-L100] could be repurposed by marking verified-but-old references as “stale”. We would compute accuracy of flagging on this test set. For extrinsic evaluation, we might measure the impact on recommendation quality: e.g., compare citation recommendation lists with vs. without stale-filtering (older vs. newer versions).

Possible automatic metrics include:

- **Outdated Citation Ratio (OCR):** the fraction of flagged citations among all references.
- **Mean Citation Age:** average publication lag of references; pipeline aim is to reduce this.
- **User-centric metrics:** in a user study, how often do flagged warnings match expert judgment that a reference is obsolete?

Datasets: Our prototype can be tested on open corpora such as S2ORC (Semantic Scholar Open Research Corpus), the PubMed Central Open Access set, or arXiv collections. For example, for each paper in a domain, feed its references through the pipeline and measure how many have a significantly newer “replacement” paper that the pipeline correctly identifies. Using domain-specific corpora (e.g., biomedical or AI conference papers) can help tune the recency thresholds.

Challenges and Limitations

Several challenges arise in automating outdated citation detection:

- **Domain Variability:** Fields differ in how fast knowledge evolves. A 5-year-old paper may be outdated in AI, but not in mathematics. Thus static age thresholds can misfire. The pipeline should ideally learn domain-specific aging patterns, but this requires classification of discipline or use of field-specific half-lives, which is a research challenge.
- **Context Sensitivity:** Not all old citations are inappropriate. Seminal works (like Newton, or foundational theory papers) remain citable despite age. Distinguishing “historically important” old references from “stale, replaceable” ones is hard. It may require analyzing the *context* of citation: e.g. if the text cites an old paper as classical groundwork, that is fine; but citing a 2005 technique to justify a modern result might be stale. This nuance is difficult to capture purely algorithmically.
- **Incomplete and Noisy Data:** Bibliographic APIs may lack some metadata or have errors. Author names can be ambiguous, and older records might have incomplete citations. We may fail to match a citation to a database entry, or misidentify a newer version. Crossref/OpenAlex cover a lot, but no source is complete. Reliance on external APIs also raises practical issues (rate limits, downtime).
- **Evolving Nomenclature:** Terminology changes over time. A topic might be cited under different keywords across decades, making it hard to match “older synonyms” to current terms via simple keyword search. Advanced NLP is needed to recognize concept equivalence.
- **False Positives/Negatives:** The pipeline might flag references as outdated when they are still relevant, leading to alert fatigue. Conversely, some outdated citations might escape detection if no clear replacement is found. Balancing sensitivity is nontrivial.
- **Evaluation Difficulties:** Creating ground truth for “outdatedness” is inherently subjective. Automated measures (like age > threshold) may not align with expert opinions about relevance. So evaluating the system will likely require user studies or crowdsourced judgments of flagged versus unflagged citations.

Despite these challenges, even partial automation can significantly assist authors and reviewers by highlighting suspiciously old references for manual check.

Research Gaps and Future Directions

Key open questions remain:

1. **Adaptive Age Modeling:** How to automatically learn what “outdated” means in each field or subfield? Future work could integrate machine learning on citation timelines to predict the current validity of a reference.

2. **Semantic Change Detection:** Develop NLP methods to detect concept drift between a citation's content and the latest state of knowledge. For instance, does the cited result still hold under new evidence? This goes beyond metadata to deep semantic analysis.
3. **User Interface and Integration:** How best to present flagged citations to researchers? Smart UI design is needed (e.g. in reference managers or writing tools) to integrate pipeline alerts without disrupting workflow.
4. **Benchmark Development:** There is a need for public datasets of papers annotated with outdated references. This would enable training and evaluation of more sophisticated models.
5. **Combination with Quality Checks:** Beyond outdatedness, future systems could jointly check for other reference issues (e.g. retracted sources [30†L138-L145] , non-accessible papers, or plagiarized content). An integrated "reference integrity" platform could cover all such concerns.
6. **Ethical and Practical Considerations:** What policies should govern automated flagging? Over-reliance on such tools could discourage citing classic works or penalize authors unfamiliar with them. Guidelines and explainability (why a citation was flagged) are important future aspects.

Conclusion

Ensuring that scholarly citations remain up-to-date is critical in the era of AI-assisted research. We have outlined a comprehensive pipeline to automatically flag outdated citations in AI-recommended literature. By leveraging open bibliographic databases (Crossref, OpenAlex) and designing metrics for citation "staleness," this pipeline can help maintain the relevance and integrity of references. Preliminary evidence from related work indicates this is both feasible and necessary: LLMs and AI tools currently propagate outdated or retracted references [13†L375-L382] [30†L138-L145] . The proposed system would integrate retrieval, temporal analysis, and expert-informed heuristics to warn users of potentially obsolete sources. Our discussion highlights both the promise and challenges of this approach, pointing to future research in adaptive aging models and semantic relevance. Ultimately, such automation will complement, not replace, human judgment—serving as a safety net that draws attention to citations warranting review. As AI continues to transform literature discovery, robust pipelines for reference validation will be essential to uphold research quality and trust [60†L92-L100] [56†L268-L271] .

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